

KITEWRIGHT CORE · DESIGN DECISIONS REGISTER

Kitewright Core design decisions register

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DOCUMENT CONTROL

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Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kitewright-core		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	O2 decided by Amish as option A (round 3 item 7A, KWC-DDR-004) and moved to Decisions made; R8 restated
0.2	2026-10-03	Amish Chadha	O1 decided by Amish as option B (round 2 item 33B, KWC-DDR-003) and moved to Decisions made; new item O2 (R8 harness change) proposed; T7 reworded for the carbon plate; cost updated
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; decisions D1 to D15 and C1 to C10 made under Amish's 2026-10-03 pre-approval; R9 core mass open for Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Safety: Decisions D3, D6 and D7 set the lithium power path, the payload retention and the load path into the frame. Any change to them reopens the build plan's safety stops.

■ Open decisions

None. Amish decided O2 (R8, the frame harness change) on 2026-10-04 as option A; see Decisions made and KWC-DDR-004. Nothing in carrying it out needs Amish.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
T1	DS-014 licence terms	Whether the connector and pinout may be reused in an open design (D12)	KWC-PRB-001
T2	DS-014 connector part numbers, pinout and current per battery pin	R1 and R4: 100 W at 6S needs 5.6 A across the battery pins	KWC-CAL-001 section C
T3	Flight controller size and mounting pattern	Damping plate holes and lid headroom (5 mm or more)	KWC-DWG-108
T4	Indexing plunger stroke (at least 9 mm) and pin length	The pin must clear the shoe when withdrawn and enter it 4.5 mm when released	KWC-DWG-105
T5	Damper hardness for the controller's mass	Vibration isolation	Build plan 3.9
T6	Power board, power module and radio footprints	Layout inside the lid; clearances checked in the model	<code>cad/src/model.py</code>
T7	M3 countersunk heads in the 2 mm carbon plate: flush seating and no pull-through (0.4 mm land under each head); bonded flush inserts as the fallback	Flush underside where the shoe slides 0.3 mm below (KWC-DDR-003)	KWC-DWG-101
T8	Telemetry band and power allowed at the first site (India: 865 to 867 MHz candidate)	Radio choice (D10)	KWC-DDR-001
T9	Bought-part masses	R9 is met by only 10 g on catalogue class values	KWC-CAL-001 section H
T10	Carbon plate laminate and price from the cutting service (3K twill, quasi-isotropic, 2 mm)	250 MPa bending and 200 MPa bearing allowables; USD 58 estimate	KWC-CAL-001 section A; <code>bom/bom.csv</code> line 1

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 3,946.80 (USD 1,053.20 under the target). Main cost drivers: the rugged ground station tablet (USD 900), the spares kit (USD 560), the telemetry radio pair (USD 400), the flight controller (USD 390), the field charger (USD 350) and the control handset (USD 250); the made parts cost about USD 150 in all, the carbon plate (USD 58) the largest of them. Savings worth trying: a tablet the team already owns if it is rated to -20 °C, a control link with built-in MAVLink telemetry in place of the separate radio pair (about USD 400), and a smaller spares kit once the first core has flown.

■ Decisions made

D1 to D15 and C1 to C10 were decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds".

Date	Decision	Decided by	Record
2026-10-03	D1: PX4 for the whole family; ArduPilot documented as the alternative	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D2: Pixhawk FMUv6X class flight controller	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D3: 18 to 60 V bus, two pack inputs, two frame outputs, AS150 connectors	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D4: one payload class for version 1, 5 kg and 100 W	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D5: plain rail, rear entry, front stops, hand-plugged DS-014 pigtail	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D6: two independent locking pins with red bands (conservative; a vibration and pull test on the built mount could relax it to one)	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D7: core hangs under the frame deck, 220 x 130 mm M4 pattern, 200 x 112 mm deck opening	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D8: optional companion computer in its own box on the frame	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D9: ColdCell reports over DroneCAN	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D10: 900 MHz class radios set per country; India 865 to 867 MHz as first candidate	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D11: three altitude parameter bands	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D12: proceed with DS-014 and confirm its licence; own open pinout as fallback	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D13: first co-design candidate to approach, a university glaciology group in Jammu and Kashmir (not agreed)	Amish (pre-approval)	KWC- DDR-001
2026-10-03	D14: log every flight; remote ID module on CAN when required		

Date	Decision	Decided by	Record
		Amish (pre-approval)	KWC-DDR-001
2026-10-03	D15: publish the hover-time model	Amish (pre-approval)	KWC-DDR-001
2026-10-03	C1 to C10: design for construction (rail, pins, stops, pigtail, clinch nuts, frame fixing, damping, lid, leads, mast)	Amish (pre-approval)	KWC-DDR-002
2026-10-03	O1 (round 2 item 33B): R9 core mass, option B. Lighter structure (carbon plate with countersunk M3 fixings in place of clinch nuts, pocketed rail bars, 12 mm pin blocks, 12 mm corner spacers, 1.5 mm lid walls) and the power leads and AS150 plugs moved to each frame's harness, with a strain-relief bar on the core: 0.99 kg (R9 met on paper), USD 6.20 more in all. Amish: "i agree with all the 46 recommendations you provided. please proceed."	Amish	KWC-DDR-003
2026-10-04	O2 (round 3 item 7A): R8 common core, option A. Accept a bench change of the frame's power harness when a core moves between Lift and Range (lid off, four 8 AWG leads desoldered and the other frame's soldered on, about 30 min); R8 restated as "same core flies Lift and Range with parameter changes and a change of the frame's power harness at the board". No hardware change; R9 stays met. Amish: "For round 3, I agree with all your proposed recommendations"	Amish	KWC-DDR-004